PV101 Day 8 Master Print Packet

Compiled packet

Day_08_PV101_Final_Review_and_Assessment_2026-07-16

Contents follow in instructional order.

PV101 Day 8 Daily Run Sheet

Instructor-facing plan for Rich or a substitute | Thursday, July 16, 2026

Day_08_PV101_Final_Review_and_Assessment_2026-07-16

Topic: Final Review and Assessment

Class setup: 6 students | 10:00 AM-3:00 PM | Calm, practical, evidence-first instruction

Teaching frame

- Main message: the final checks whether students can apply PV101 fundamentals safely and clearly.
- The final is a traditional graded assessment: 25 questions, one point each.
- Do not coach answers during the final. Clarify directions or unreadable wording only.
- PV101 ends today with reflection, evidence collection, and a clean instructional closeout—no capstone or certification exercise.

Substitute quick start

- [ ] Read the full schedule before students arrive.
- [ ] Locate emergency disconnects, first-aid supplies, calculator policy, and secure test storage.
- [ ] Print student materials and keep answer keys/instructor notes separate.
- [ ] Write the day goal, schedule, and stop-work rule on the board.
- [ ] If uncertain about equipment or assessment handling, use the reduced-scope path and contact program leadership.

Learning targets

- Recall and apply core ideas from all 12 assigned chapters.
- Use formulas, diagrams, sequencing, and written reasoning accurately.
- Choose safe verification and troubleshooting steps instead of guessing.
- Complete a traditional graded 25-question assessment independently.
- Reflect on PV101 growth and leave materials organized and secure.

Materials

- Readiness quiz and key
- Day 8 slide deck and whiteboard outline
- 25-question final tests and secured answer key
- Calculators and scratch paper as allowed
- Reflection sheet and evidence checklist
- Folders/envelopes for completed graded work

Daily schedule

Time	Block	Instructor notes
10:00-10:10	Arrival, attendance, check-in	Students receive readiness quiz only. Final tests remain secured.
10:10-10:20	Day goal + test expectations	Explain the day: review, demonstrate individual understanding, reflect, and close PV101.
10:20-10:35	Morning readiness quiz	Short five-question check distinct from the graded final.
10:35-10:55	Readiness review	Correct together. Use results to choose which misconceptions need the most review.
10:55-11:20	Whiteboard review	Sort concepts into energy path, measurement, workmanship, protection, troubleshooting, and sizing.
11:20-11:30	Break	Students reset. Instructor keeps final tests and answer key controlled.
11:30-12:00	Slides: cumulative review	Review formulas, meter discipline, protection, series/parallel, troubleshooting evidence, and sizing steps.
12:00-12:20	Practice questions	Work examples that are similar in skill but not identical to final-test questions.
12:20-12:30	Final instructions	Explain 25 questions, one point each, calculator policy, quiet conditions, and how to request clarification.
12:30-1:00	Lunch	Final tests remain secured. No discussion of test content.
1:00-2:10	PV101 final assessment	Traditional graded 25-question cumulative test. Instructor monitors without coaching answers.

2:10-2:20	Collect + verify materials	Count every test and scratch page. Store answer key separately.
2:20-2:35	Course reflection	Students reflect on verification, growth, remaining practice needs, and safe-work habits.
2:35-2:45	PV101 closeout discussion	Name the skills completed and state simply that PV101 is complete. No capstone or project presentation.
2:45-2:55	Evidence + Classroom upload	Submit reflection and any required learning evidence; instructor completes evidence checklist.
2:55-3:00	Cleanup + dismissal	Room reset, materials counted, student papers secured, PV101 formally closed.

Safety and instructional stop conditions

- Final tests or answer keys are unaccounted for.
- Students discuss answers during the individual test.
- A question appears to require content not taught; note it for review without coaching an answer.
- A student becomes overwhelmed; provide a quiet reset while preserving test integrity.
- Assessment materials are photographed or uploaded without authorization.

Reduced-scope backup

Day 7: if calculators or printing fail, complete one class example on the whiteboard and have students show units and reasonableness on notebook paper. Day 8: if final-test printing fails, run structured review and reflection; do not improvise a different final from memory. Reschedule the secured assessment.

Early finishers

Day 7: check every unit, redo one calculation independently, and write one sentence explaining each sizing result. Day 8: quietly review answers, then begin the reflection only after turning in the final.

Instructor notes after class

What landed:

_____

_____

What confused students:

_____

_____

Equipment/material issues:

_____

_____

Carry-forward:

_____

_____

PV101 Day 8 Morning Quiz

Student copy | 10-15 minutes

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Name: _____ Date: _____

Answer briefly. Show work and units where needed.

1. A 24 V training load draws 2 A. What power does it use?

_____

_____

2. A 40 W device operates for 2 hours. What daily energy does it use?

_____

_____

3. Before testing continuity across a fuse, name the required energy state and meter function.

_____

_____

4. A wire label says polarity is correct. What evidence should verify the label?

_____

_____

5. Why write the expected value and unit before taking a measurement?

_____

_____

PV101 Day 8 Morning Quiz Answer Key

Instructor copy

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1. A 24 V training load draws 2 A. What power does it use?
48 W; W = V x A = 24 x 2.

2. A 40 W device operates for 2 hours. What daily energy does it use?
80 Wh; Wh = W x h = 40 x 2.

3. Before testing continuity across a fuse, name the required energy state and meter function.
Circuit de-energized and verified; meter on continuity/ohms with correct leads and ports.

4. A wire label says polarity is correct. What evidence should verify the label?
A correctly configured voltage measurement at documented points with the observed polarity compared to the expected polarity.

5. Why write the expected value and unit before taking a measurement?
It helps select the correct function/range and makes the observed result useful evidence instead of an unexplained number.

PV101 Day 8 Whiteboard Outline

Instructor-facing

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Board before students enter

- PV101 Day 8 | Review -> Final -> Reflection -> Closeout
- 25 questions | 25 points
- Safety overrides speed. Units count. Evidence beats labels.

Six review columns

- Energy path
- Electricity + formulas
- Meter verification
- Workmanship + serviceability
- Protection + troubleshooting
- Loads + sizing

Misconceptions to correct

- Wire colors and labels are hints—not proof.
- Continuity requires a de-energized circuit.
- A replaced fuse is not a verified repair.
- Series adds voltage; parallel adds current.
- A sizing answer without units or assumptions is incomplete.

Final-test boundaries

- Independent work.
- Clarify directions only.
- No answer coaching.
- Collect every test and scratch page.

PV101 Day 8 Review Worksheet

Student copy | Use before the final

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Name: _____ Date: _____

Formula and units check

Write the power formula: _____

Calculate 24 W used for 3 h: _____ Wh

Convert 0.5 kW to watts: _____ W

In series, _____ adds. In parallel, _____ adds.

Verification chain

Put in order: evidence, symptom, conclusion, test.
Answer: _____

System path

Draw and label the basic PV-to-load energy path. Mark one fuse, one disconnect, and one meter point.

_____

_____

_____

Common misconceptions to correct

1. Wire color proves polarity. Correct / Incorrect: _____
Correction: _____

2. Continuity is tested on an energized circuit. Correct / Incorrect: _____
Correction: _____

3. Replacing a fuse proves the fault is repaired. Correct / Incorrect: _____
Correction: _____

4. Loads are the starting point for sizing. Correct / Incorrect: _____
Correction: _____

5. Cleanup is part of technical work. Correct / Incorrect: _____
Correction: _____

PV101 Day 8 Student Worksheet Answer Key

Instructor copy

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Formula and units check

- Power formula: W = V x A.
- 24 W x 3 h = 72 Wh.
- 0.5 kW x 1,000 W/kW = 500 W.
- In series, voltage adds. In matched parallel paths, current adds.

Verification chain

Symptom -> test -> evidence -> conclusion. Accept equivalent wording that keeps the observation before the test and the conclusion after evidence.

System path

Accept PV -> controller -> battery -> protected inverter/DC load path with a complete return, one source-side fuse, one disconnect, and one logical meter point.

Misconceptions

Statement	Answer	Correction
Wire color proves polarity.	Incorrect	Color is a clue; verify polarity with labels, tracing, and the correct meter setup.
Continuity is tested on an energized circuit.	Incorrect	Continuity/ohms testing requires the circuit to be de-energized.
Replacing a fuse proves the fault is repaired.	Incorrect	Find the cause and verify the restored known-good state.
Loads are the starting point for sizing.	Correct	Document watts, hours, simultaneous loads, and assumptions before sizing.
Cleanup is part of technical work.	Correct	A professional closeout includes safe state, material count, records, and room reset.

PV101 Day 8 Guided-Practice Guide

Student-facing

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Name: _____ Date: _____

Purpose

Provide the student-facing sequence for review, final assessment, reflection, and PV101 closeout.

Materials

- Readiness quiz
- Review worksheet
- Calculator and pencil
- Final assessment
- Reflection questions

Student sequence

- Complete the readiness quiz.
- Participate in cumulative review.
- Use lunch as a test-content break.
- Complete the final independently.
- Turn in the final before starting reflection.
- Submit reflection and leave the room reset.

Test rules

- One point per question.
- Show units on calculations.
- Ask only for clarification of directions.
- Do not discuss or photograph test content.

Cleanup

- Return calculators.
- Submit every test and scratch page.
- Upload/submit reflection as directed.
- Clear tables and wait for dismissal.

PV101 Day 8 Reflection Questions

Student copy

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Name: _____ Date: _____

1. What PV101 skill improved most for you?

_____

_____

2. Which safety or verification rule will you remember?

_____

_____

3. Which topic still needs practice?

_____

_____

4. Describe one time evidence changed your conclusion.

_____

_____

5. What does professional cleanup/documentation mean to you?

_____

_____

PV101 Day 8 Instructor Notes

Instructor-facing

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Before class

- Print and separate student/instructor copies.
- Verify the 10:00-3:00 schedule and room setup.
- Keep answer keys controlled.

During class

- Use missed quiz items as teaching feedback.
- Require units and evidence.
- Protect student dignity; correct privately and calmly.

After class

- Record what landed and what needs follow-up.
- Secure graded work.
- Complete the evidence checklist and Daily Summary.

PV101 Day 8 Evidence Checklist

Instructor copy

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- [ ] Readiness quiz completed
- [ ] Cumulative review completed
- [ ] Final test issued and counted
- [ ] All 25 questions present
- [ ] Final test returned and secured
- [ ] Raw score recorded out of 25
- [ ] Reflection submitted
- [ ] Classroom evidence submitted
- [ ] PV101 closeout completed
- [ ] Room and materials reset

Instructor initials: _____ Notes: _____

PV101 Final Assessment

Student copy | 25 questions | 25 points

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Name: _____ Date: _____ Score: _____ / 25

Directions: Work independently. Circle one answer for multiple choice. Show formulas and units for calculations. Write clearly for explanations. Each question is worth one point.

1. Photovoltaic (PV) equipment converts which resource directly into electricity?

- A. Moving air
- B. Light
- C. Flowing water
- D. Fuel vapor

2. Which application most clearly needs energy storage to operate when sunlight is unavailable?

- A. A direct daytime PV fan
- B. A grid-tied array with no backup requirement
- C. An off-grid cabin used at night
- D. A module demonstration with no load

3. Before changing wiring on a training PV system, the safest first action is to:

- A. Work faster
- B. Isolate energy sources and verify the state
- C. Trust the switch label
- D. Replace the fuse

4. A student is uncertain about polarity. What should happen?

- A. Connect briefly and watch
- B. Ask a classmate to guess
- C. Stop and verify with the correct meter setup
- D. Reverse both conductors

5. Which statement is correct?

- A. Voltage is electrical pressure and current is flow
- B. Current is measured in volts
- C. Power is measured in ohms
- D. Resistance creates energy

6. A 12 V fan draws 3 A. Calculate its power in watts. Show the formula.

_____

_____

_____

7. Peak sun hours are used mainly to estimate:

- A. Daily equivalent full-sun energy
- B. Wire color
- C. Battery terminal size
- D. AC frequency

8. What usually happens to PV output when a module is heavily shaded?

- A. Output improves
- B. Output is reduced
- C. AC voltage doubles
- D. Battery capacity increases

9. A PV module exposed to light should be treated as:

- A. Always de-energized
- B. A possible DC voltage source
- C. An AC-only device
- D. A fuse

10. Read this connection diagram: [Module 1] -- series -- [Module 2]. Which electrical quantity normally adds in this matched series connection?

_____

_____

_____

11. What is the main job of a charge controller in a battery-based PV system?

- A. Convert DC to AC
- B. Regulate charging to protect the battery
- C. Create sunlight
- D. Replace overcurrent protection

12. What is the main job of an inverter?

- A. Convert DC to AC
- B. Store energy
- C. Measure resistance
- D. Increase sun hours

13. Three 10 W lights operate 4 hours per day. Calculate their total daily energy use.

_____

_____

_____

14. What should be completed before selecting battery, array, or inverter size?

- A. A load assessment
- B. A wire-color chart
- C. A tool inventory only
- D. A final photograph

15. Why is a fuse placed close to a battery source?

- A. To increase voltage
- B. To protect as much conductor as practical
- C. To make the inverter quieter
- D. To eliminate shutdown steps

16. A long undersized conductor carrying current is most likely to cause:

- A. Lower voltage drop
- B. Voltage drop and heating
- C. More sun hours
- D. Increased battery capacity

17. Which is a reasonable advantage of a ground-mounted array?

- A. It always costs less
- B. It can be easier to access and orient
- C. It cannot be shaded
- D. It needs no structure

18. A roof area is shaded from 1:00-4:00 PM, while an open ground area has clear solar access. Circle the better conceptual location and write one reason.

_____

_____

_____

19. After observing a fault symptom, the best next step is to:

- A. Replace several parts
- B. Choose one test that can separate likely causes
- C. Bypass protection
- D. Trust the label

20. Explain why replacing a blown fuse is not enough to prove the system is repaired.

_____

_____

_____

_____

_____

21. In a permitting context, the AHJ is the organization or official that:

- A. Guarantees system output
- B. Sets the price of every module
- C. Enforces the applicable local requirements and approvals
- D. Replaces the serving utility

22. A current, documented rebate in a training example covers $500 of a $2,500 cost. Calculate the remaining cost before any other fees or credits.

_____

_____

_____

23. After completing a set of practice questions, explain why reviewing the missed questions is more useful than recording only the score.

_____

_____

_____

_____

_____

24. A cumulative practice question asks you to convert 0.75 kW to watts. Show the conversion.

_____

_____

_____

25. A small PV training system does not power its load. The panel is in light, the controller display is off, and a label says the battery fuse is good. Describe the safe first three actions/tests and the evidence you would document.

_____

_____

_____

_____

_____

PV101 Final Assessment Answer Key

Instructor copy | 25 questions | 25 points

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Q	Chapter	Type	Answer / acceptable evidence
1	1	MC	B. Light
2	1	MC	C. An off-grid cabin used at night
3	2	MC	B. Isolate energy sources and verify the state
4	2	MC	C. Stop and verify with the correct meter setup
5	3	MC	A. Voltage is electrical pressure and current is flow
6	3	CALC	36 W; W = V x A = 12 x 3.
7	4	MC	A. Daily equivalent full-sun energy
8	4	MC	B. Output is reduced
9	5	MC	B. A possible DC voltage source
10	5	DIAGRAM	Voltage adds; current remains limited by the matched module/string current.
11	6	MC	B. Regulate charging to protect the battery
12	6	MC	A. Convert DC to AC
13	7	CALC	120 Wh/day; 3 x 10 W x 4 h.

14	7	MC	A. A load assessment
15	8	MC	B. To protect as much conductor as practical
16	8	MC	B. Voltage drop and heating
17	9	MC	B. It can be easier to access and orient
18	9	DIAGRAM	Ground area; it has better documented solar access. Accept reason tied to shading/access, with acknowledgment that full site review is still required.
19	10	MC	B. Choose one test that can separate likely causes
20	10	SHORT	The cause and conductor/system condition must be checked, then the known-good state must be verified with repeat measurements or functional testing.
21	11	MC	C. Enforces the applicable local requirements and approvals
22	11	CALC	$2,000; $2,500 - $500 = $2,000.
23	12	SHORT	Reviewing misses identifies the specific concept, calculation, unit, or safety reasoning that needs correction and turns the practice set into a learning tool.
24	12	CALC	750 W; 0.75 kW x 1,000 W/kW.

25	Cumulative	SCENARIO	Accept: stop and isolate energy; inspect/trace the actual path and protection; verify meter setup; with the circuit de-energized test fuse continuity or measure voltage at logical points after safe re-energization; record expected vs actual readings, conclusion, repair, and restoration check. Must not bypass protection or trust the label alone.

Scoring

Award one point per question. For calculations, require the correct result with an appropriate unit; accept equivalent work. For written items, accept equivalent safe reasoning that includes the listed evidence. Record the raw score out of 25.